Towards Evaluating the Robustness of Neural Networks

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1 Introduction

In recent years, the performance of deep neural networks (DNNs) has increased significantly. As a result, they are now used in numerous areas of application, from image recognition and speech processing to safety-critical domains. However, despite these successes, DNNs have fundamental weaknesses, in particular, their high susceptibility to small input changes [CW17]. While such errors can occur unintentionally, the question arises as to what happens when these weaknesses are exploited deliberately and how we can protect ourselves against this [CW17].

The importance of this topic is obvious: in security-related fields of application, manipulative interference with data could have catastrophic consequences. From the miscontrol of autonomous vehicles to attacks on critical infrastructure [CW17]. In addition, the evaluation and improvement of the robustness of neural networks is becoming increasingly important against the backdrop of growing regulatory requirements for artificial intelligence. The development of robust AI that functions reliably even under attack is, therefore, central to the acceptance and safe use of modern AI technologies.

At the same time, adversarial robustness is no longer a purely academic concern but increasingly a prerequisite for deploying AI systems in safety- and mission-critical environments, where failures can entail legal, financial, and reputational risks for operators and manufacturers [CW17; Sze+13]. Consequently, there is a growing need for systematic evaluation frameworks and standardized benchmarks that make robustness claims comparable across models, datasets, and application domains and that support emerging regulatory auditing practices for AI systems [CW17; Sze+13].

Szegedy et al. [Sze+13] first drew attention to this problem by introducing the concept of adversarial examples: small perturbations, often barely perceptible to humans, that can cause neural networks to make incorrect or even deliberately manipulable classifications. Szegedy et al. [Sze+13] revealed this vulnerability through optimization-based attacks using L-BFGS. They demonstrated this vulnerability using an optimization method based on L-BFGS, thereby highlighting fundamental security and reliability issues in the use of neural networks in real-world security-relevant environments.

Subsequently, numerous attack and defense methods were developed. Goodfellow, Shlens, and Szegedy [GSS15] presented an efficient attack approach with the Fast Gradient Sign Method (FGSM) and argued in their linearity hypothesis that the high linearizability of modern models structurally favors their vulnerability. Papernot et al. [Pap+16] , on the other hand, presented the approach of defensive distillation as a possible defense strategy. By training a model with "softer" classification probabilities from a previously trained model, the decision boundary should be smoothed and thus become more robust against small disturbances. However, it became apparent early on, based in part on the linearity hypothesis, that this method misses its mark. This was ultimately confirmed by the work of Carlini and Wagner [CW17], who succeeded in completely circumventing defensive distillation.

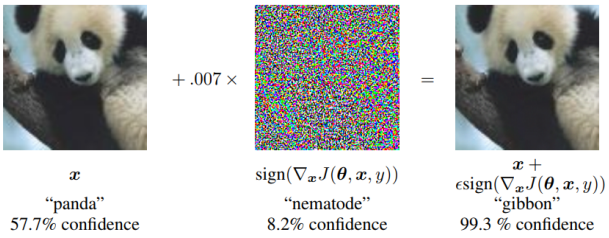


Figure 1: Example of an adversarial image generated with the Fast Gradient Sign Method (FGSM). A small perturbation proportional (.007) to the sign of the input gradient is added to the original image, which is imperceptible to humans but causes the classifier to change its prediction from panda to gibbon with high confidence [GSS15].

Research in the field of adversarial robustness thus faces a number of fundamental challenges. However, new defense approaches are needed that not only work against specific attacks, but are also as generalizable and reliable as possible. This requires a clear understanding of threat models, attack classes, and security aspects, as conveyed by Chakraborty et al. [Cha+21] and Zhou et al. [Zho+22]. These works not only illustrate the diversity of existing attacks, from gradient-based to heuristic to black-box approaches, but also highlight key challenges such as the transferability of adversarial examples, the lack of generalizability of many defenses, and the inherent trade-off between robustness and model performance.

Against this background, this seminar paper focuses on analyzing the robustness of neural networks based on the work of Carlini and Wagner [CW17]. The aim is to understand how their attacks work, why they are so effective against previously established defense mechanisms, and what implications this has for the development of future robust and trustworthy machine learning systems. In particular, this work aims to systematically trace the optimization-based attack formulations by Carlini and Wagner [CW17] and to situate their contribution within the broader landscape of adversarial robustness evaluation. Furthermore, the seminar paper illustrates why naive, purely heuristic attack evaluations often lead to overly optimistic robustness claims and how stronger, optimization-based attacks can correct this picture and reveal hidden failure modes of existing defenses.

This seminar paper is structured as follows. Section 2 provides essential background on neural network architectures, adversarial examples, L_p-norm threat models, and classical attacks. Section 3 analyzes the limitations of heuristic verification and gradient masking in defensive distillation. The core analysis in Section 4 systematically reconstructs the optimization-based attacks of Carlini and Wagner [CW17] under L_0, L_2, and L_∞ norms, their unified formulation, and evaluation results that expose distillation's failure. Finally, Section 5 situates these contributions within related verification approaches, while Section 6 discusses implications for modern robustness evaluation.

2 Background

The following subsections build the conceptual foundation for the Carlini and Wagner-attacks: starting from NN-architectures, over adversarial examples and threat models, to classical attacks and defenses.

2.1 Foundations: Functioning of NNs/DNNs and their vulnerable properties

Artificial neural networks (NNs) and deep neural networks (DNNs) are inspired by biological neural systems, but are best understood as mathematical function approximators composed of multiple interconnected layers [LBH15]. Their primary goal is to learn complex, non-linear mappings from input data to output predictions by hierarchically transforming representations across layers.

A DNN consists of an input layer, one or more hidden layers, and an output layer. Each layer comprises a set of neurons, where each neuron applies a weighted linear transformation followed by a non-linear activation function. Formally, each layer computes a function of the form:

$$h_l : x_{l-1} \to g(W_l x_{l-1} + b_l)$$

where W_l and b_l denote the weights and biases of layer l, and g is an activation function such as ReLU or sigmoid [Mon+14].

The weighted connections between consecutive layers determine how information is propagated through the network [GSS15]. During training, these weights are adjusted using gradient-based optimization methods, enabling the network to learn task-specific patterns for classification or regression [GSS15]. Although this architecture allows DNNs to achieve high performance in complex tasks, it also introduces properties such as high dimensionality, piecewise linear behavior, and differentiability [GSS15]. These characteristics play a crucial role in the vulnerability of DNNs to adversarial attacks, as they allow small, carefully crafted perturbations to propagate through the network and significantly alter the final prediction [Mon+14; GSS15]. In the context of adversarial

robustness, this functional viewpoint is particularly useful because it allows one to relate architectural choices and training objectives directly to properties of the decision boundary, such as smoothness, curvature, and local linearity. Having established the functional architecture of DNNs and their inherent vulnerabilities, the next subsection introduces adversarial examples as concrete manifestations of these weaknesses.

2.2 Definition and discovery of adversarial examples (Szegedy)

Adversarial examples are inputs that have been intentionally perturbed to cause a machine learning model to output an incorrect prediction, often with high confidence [CW17; SPV19]. The perturbation is constrained to be very small in a suitable norm so that the modified input remains (almost) indistinguishable from the original to a human observer [Sze+13].

Szegedy et al. [Sze+13] first discovered that state-of-the-art image classifiers are vulnerable to such adversarial examples, i.e., inputs obtained by adding tiny, carefully chosen perturbations to correctly classified images that lead the network to misclassify them while remaining visually unchanged for humans. Later work showed that crafting adversarial examples can be seen as an optimization problem and that even simple gradient-based methods efficiently find such inputs for many modern neural networks [Sze+13; GSS15; CW17]. From an evaluation perspective, adversarial examples serve as a stress test for neural networks, revealing worst-case vulnerabilities that are not visible under standard test accuracy and thus complement conventional generalization metrics [CW17; SPV19]. In the setting of targeted adversarial examples, we distinguish three ways to choose the target class t [CW17]:

- **Average case:** Choose a target label $t \neq C(x)$ uniformly at random and construct x' such that $C(x') = t$.
- **Best case:** For all $t \neq C(x)$, attempt to find an adversarial example and report the target class for which the attack requires the smallest perturbation.
- **Worst case:** Analogously, report the target class $t \neq C(x)$ for which a successful attack requires the largest perturbation.

Formally, let $C : \mathbb{R}^n \to \{1, \ldots, K\}$ be a classifier and let $x \in \mathbb{R}^n$ be an input with true label y such that $C(x) = y$. An (untargeted) adversarial example x' for x under a norm $\|\cdot\|_p$ and budget $\varepsilon > 0$ satisfies

$$C(x') \neq y \quad \text{and} \quad \|x' - x\|_p \leq \varepsilon.$$

In the targeted setting with target class $t \neq y$, an adversarial example x' instead satisfies

$$C(x') = t \quad \text{and} \quad \|x' - x\|_p \leq \varepsilon.$$

Untargeted attacks are generally easier to construct, as they only require the input to cross any decision boundary and induce misclassifications into an arbitrary incorrect class. In contrast, targeted attacks impose a stricter constraint by forcing the model to assign a specific, attacker-chosen target label, which typically requires more structured perturbations and stronger control over the decision boundary. Consequently, targeted attacks are often regarded as a more informative indicator of model vulnerability and are frequently used as a benchmark in robustness evaluations [Sze+13; GSS15; CW17].

An important property of adversarial examples is their transferability: adversarial inputs crafted for one model often remain effective against other models, even with different architectures or training data [CW17]. Transferability is often attributed to shared high-level feature representations and similarly shaped decision boundaries learned by different models on the same task, which causes adversarial perturbations to generalize across architectures [GSS15]. This phenomenon enables black-box attacks and further highlights the fundamental nature of adversarial vulnerability in modern neural networks [CW17]. Understanding both targeted and transfer-based attacks is crucial for designing robust training schemes and for evaluating defenses under realistic black-box threat models [GSS15]. With adversarial examples defined, we now examine the key distance metrics (L_p-norms) that constrain perturbations and define realistic threat models.

2.3 Distance Metrics and Threat Models

To quantify similarity, we are using three L_p norms that measure distances between two vectors $x, y \in \mathbb{R}^n$:

$$\mathbf{L_0}: \quad d(x,y) = \|x-y\|_0 = \big|\{\, i \in \{1,\dots,n\} : x_i - y_i \neq 0 \,\}\big|$$

$$\mathbf{L_2}: \quad d(x,y) = \|x-y\|_2 = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$$

$$\mathbf{L_\infty}: \quad d_\infty(x,y) = \|x-y\|_\infty = \max_{1\le i\le n} |x_i - y_i|$$

Each of these norms has different aspects and use cases in the analysis of perturbations. The so-called L_0 "norm" (strictly not a true norm) directly controls how many coordinates are allowed to change and is therefore well suited for modeling sparse attacks, where the adversary aims to modify as few input dimensions as possible (e.g., changing only a small set of pixels or features) [CW17]. optimization under L_0 constraints is typically combinatorial and non-differentiable, which makes attack generation and robustness evaluation computationally demanding [Cha+21; KV20]. The L_2 norm corresponds to the Euclidean distance and evaluates the overall magnitude of all differences and is mathematically convenient due to its smoothness and rotational invariance [CW17]. It is widely used in robustness benchmarks and theoretical analyses, as it allows gradient-based optimization and often leads to well-behaved optimization landscapes [CW17]. But L_2-bounded perturbations can be spatially diffuse. Many small changes may sum to a moderate L_2 distance, even though humans might perceive them as either negligible or in some cases, more noticeable than the norm value suggests [Cha+21]. The L_∞ norm measures the largest deviation in any coordinate and is therefore natural for modeling uniform, small-magnitude perturbations such as slight pixel-wise noise or bounded sensor noise [CW17]. This makes L_∞ -balls a common choice in adversarial training and standardized robustness evaluations, where one assumes that no single input dimension can be changed beyond a fixed tolerance [KV20]. A limitation of the L_∞ constraint is that it ignores how many coordinates are changed: many small but widespread perturbations can still be allowed, even if they look clearly unnatural to a human observer [KV20].

In practice, the choice of norm depends on the underlying application and the threat model. For example, L_∞-bounded attacks model scenarios like subtle sensor noise in autonomous driving, where no single pixel can deviate beyond a small tolerance (e.g., $8/255$), making them standard for ϵ-bounded adversarial training benchmarks [CW17]. L_2-bounded attacks suit theoretical analyses of global Gaussian perturbations or image compression artifacts, as their smoothness enables stable gradient descent and quantifies total energy of changes across the image [Cha+21]. L_0-based constraints capture realistic sparse threats, such as targeted sticker attacks on traffic signs or malware altering few binary features in network packets, prioritizing minimal edit distance over uniform noise [KV20]. As a result, different norms highlight different aspects of model vulnerability and can lead to substantially different conclusions about robustness [CW17; Cha+21]. Consequently, selecting an L_p norm is not merely a technical detail but an integral part of the threat model, as it encodes which perturbations are considered feasible and which types of attacks are regarded as realistic in a given application. These distinctions between L_0, L_2, and L_∞ threat models foreshadow the core contribution of Carlini and Wagner [CW17], whose attacks are explicitly tailored to each norm and thereby expose how robustness conclusions can change dramatically with the chosen distance metric.

In the context of adversarial machine learning, the choice of distance metric is closely tied to the threat model [CW17; Cha+21]. A threat model specifies the attacker's capabilities and constraints. We distinguish between two main types of attacks: white-box and black-box [CW17]. They differ in the amount of information the attacker has, in a white-box setting, the adversary has full access to the model f, including the architecture, the training algorithm, the data distribution μ, and the learned parameters θ. This information is used to analyze regions of high error and to construct adversarial examples that are constrained in an L_p-ball around a given input [Cha+21]. Black-box

attacks assume that the attacker only has access to the model (inputs and outputs) and possibly some side information about the task, but not about the internal parameters or gradients [Cha+21]. In this work, we focus exclusively on white-box attacks and therefore do not consider black-box scenarios in more detail.

Beyond the white-box and black-box distinction, adversarial examples are also studied with respect to transferability between models and between digital and physical domains. Physical attacks typically require robustness under real-world transformations (e.g., printing, viewing angle, illumination), which is not fully captured by simple L_p-constraints in pixel space [CW17]. These threat models form the basis for classic attacks that provide early algorithmic instantiations, but also highlight limitations that Carlini and Wagner addresses.

2.4 Classical Adversarial Attacks Before Carlini and Wagner

Early adversarial attack methods mainly relied on gradient-based heuristics and were typically defined under simple L_p-constraints [CW17]. The Fast Gradient Sign Method (FGSM) perturbs the input in the direction of the sign of the loss gradient under an L_∞constraint, while iterative variants (e.g. I-FGSM or Projected Gradient Descent (PGD)) apply this update multiple times to generate stronger adversarial examples [GSS15; CW17]. These methods are computationally cheap and can quickly produce untargeted or targeted perturbations, which made them a standard tool for early robustness evaluations [GSS15; CW17]. More advanced heuristic approaches include the Jacobian-based Saliency Map Attack (JSMA), which selectively modifies a small set of input features and is often associated with an L_0 threat model, and DeepFool, which iteratively estimates the minimal perturbation required to cross the decision boundary under a local linear approximation in L_2-space [Pap+16; CW17]. Both attacks aim at reducing the perturbation size compared to simple one-step methods, but still rely on local gradient information and approximations [Pap+16]. As a consequence, early attack methods provided only crude upper bounds on robustness: they often produce relatively large or perceptible distortions, do not explicitly optimize for minimal L_p-distortion under a given misclassification constraint, and may fail or become unreliable against models that employ robustness-enhancing techniques such as gradient masking or defensive distillation [CW17]. These shortcomings motivated the optimization-based framework of Carlini and Wagner [CW17], which explicitly formulates adversarial example generation as a constrained optimization problem over different L_p norms, directly minimizes perturbation size, and remains effective even in the presence of gradient masking [CW17]. Finally, we discuss heuristic verification and defensive distillation, whose evaluation pitfalls motivate C&W's stronger approach (detailed in Sec. 3).

2.5 Heuristic Verification and Defensive distillation

A heuristic method is an important approach that aims to find a useful solution by trying different actions or approximations, without providing formal guarantees [RN16]. In the context of adversarial robustness, heuristic verification evaluates the robustness of neural networks by applying specific adversarial attack methods and interpreting the failure of these attacks as evidence of robustness. However, such evaluations crucially depend on the strength and suitability of the chosen attacks and may therefore lead to false conclusions, especially in the presence of gradient masking or obfuscated decision boundaries [Bas+17].

To reason more formally about the robustness of a neural network, one can aim to establish either lower or upper bounds on its robustness. A *lower bound* guarantees that no admissible adversarial example exists within a given perturbation radius, which provides a certified minimum level of robustness. In contrast, an *upper bound* is obtained by successfully finding an adversarial example within a certain distance, showing that the model is not robust beyond that radius. Heuristic attack-based evaluations typically provide such upper bounds, but without any guarantee that they are tight [CW17; Bas+17].

Defensive distillation, introduced by Papernot et al. [Pap+16], is an early heuristic defense against adversarial attacks that adapts the concept of knowledge distillation to improve model robustness. The method retrains a neural network using softened output probabilities produced by a previously trained teacher model (via an increased softmax temperature), thereby aiming to reduce

sensitivity to small input perturbations by smoothing the decision boundaries [Pap+16]. While defensive distillation was initially reported to substantially increase robustness against gradient-based attacks, subsequent work by Carlini and Wagner [CW17] demonstrated that this apparent robustness is largely due to gradient masking rather than genuine resistance to adversarial examples. As a result, defensive distillation is now considered a paradigmatic example of how heuristic defenses can lead to overly optimistic robustness assessments when evaluated using non-adaptive attacks [CW17].

As later work has shown, many of these defense mechanisms rely on heuristic verification and may give a false sense of robustness; we return to this point in Section 3.

2.6 Linearity hypothesis and vulnerable properties

The linearity hypothesis plays a central role in research on the robustness of neural networks [CW17; SPV19]. It states that many neural networks, especially deep models with piecewise linear activation functions, behave approximately linearly in their response to small perturbations of the input [SPV19]. In high-dimensional input spaces, even tiny spaces, carefully aligned changes along the gradient direction can accumulate across many coordinates and thus induce a large change in the output, while remaining almost imperceptible in each individual dimension. This perspective helps explain why neural networks can be extremely sensitive to small, adversarially chosen perturbations, despite performing well under natural noise [GSS15].

Vulnerable properties of neural networks, such as unintuitive and hard-to-predict responses to manipulated inputs, have therefore been extensively studied in the literature. Gradient-based attacks in particular exploit locally linear decision surfaces: by following the sign or direction of the loss gradient, an adversary can construct perturbations that reliably push the input across the decision boundary [GSS15]. This mechanism is closely tied to the linearity hypothesis and shows how the combination of high dimensionality and local linear behavior can systematically be used to generate adversarial examples with small norm-constrained perturbations [SPV19].

Moreover, the linearity viewpoint also offers a partial explanation for the transferability of adversarial examples between different models [SPV19]. If multiple networks trained on the same task learn similar high-level directions in feature space, then adversarial perturbations aligned with these directions can fool several models at once, even when their architectures or training procedures differ [SPV19]. Transferability is particularly relevant in black-box scenarios, where attackers exploit a surrogate model to craft perturbations that generalize to an unknown target model [SPV19].

At the same time, the linearity hypothesis does not fully explain all observed phenomena and has important limitations [SPV19]. Several works point out that geometric properties of the data manifold, the curvature and positioning of decision boundaries, as well as the structure of learned representations also play a crucial role in adversarial vulnerability [SPV19]. Consequently, linearity should be seen as one important component in a broader set of explanations rather than a complete theory. Nonetheless, it provides an intuitive and analytically tractable framework that has strongly influenced the design and analysis of gradient-based attacks and defenses, and it continues to serve as a useful starting point for understanding why modern neural networks are so vulnerable to adversarial perturbations [CW17; GSS15; SPV19].

3 Heuristic Verification

Heuristic verification refers to the empirical evaluation of adversarial robustness by applying specific adversarial attack algorithms to a trained model and interpreting the failure of these attacks as evidence of robustness [Bas+17]. This approach is widely used in practice because it is scalable and can be applied to modern deep neural networks with millions of parameters. However, heuristic verification does not provide formal guarantees: the inability of a particular attack to find an adversarial example does not imply that no such example exists within the specified perturbation region [Bas+17].

A fundamental limitation of heuristic verification is its strong dependence on the chosen attack method. Early robustness evaluations often relied on simple gradient-based attacks such as FGSM

or JSMA [GSS15; Pap+16]. While these methods are computationally efficient, they explore only a limited portion of the adversarial space and may fail even when adversarial examples exist. As a result, robustness can be substantially overestimated if the employed attacks are too weak or not well aligned with the defense mechanism [Cha+21].

One of the most prominent failure modes of heuristic verification is *gradient masking* (also referred to as obfuscated gradients). In this case, a defense alters the loss landscape such that gradients become uninformative, unstable, or numerically small, causing standard gradient-based attacks to fail without actually increasing true robustness [CW17]. Defensive distillation is a paradigmatic example of this phenomenon. By training a model using softened output probabilities, defensive distillation modifies the softmax behavior and suppresses gradients when attacks are formulated directly on the output probabilities [Pap+16]. However, as demonstrated by Carlini and Wagner [CW17], adversarial examples still exist and can be found reliably when attacks are reformulated to operate on the logits and optimized explicitly [CW17]. This shows that the apparent robustness of defensively distilled networks is largely an artifact of the evaluation procedure rather than a genuine improvement in security.

Similar issues arise in other heuristic defenses that rely on non-differentiable preprocessing steps, randomized transformations, or numerical instabilities. While such techniques may break specific attack implementations, they often remain vulnerable to adaptive attacks that approximate gradients or optimize alternative objective functions [Cha+21]. Consequently, robustness claims based solely on heuristic verification must be interpreted with caution, especially in security-critical applications.

Despite these limitations, heuristic verification remains an important component of robustness assessment. It provides practical upper bounds on robustness by exhibiting concrete adversarial examples and is often the only feasible option for large-scale networks, for which formal verification is computationally intractable [Bas+17]. To improve reliability, several best practices have emerged: attacks should be adaptive to the defense, multiple L_p threat models should be considered, optimization-based attacks should be preferred over single-step heuristics, and evaluation results should include transparent reporting of attack success rates, achieved distortion, and failure cases [CW17; Cha+21].

In contrast to heuristic verification, formal and probabilistic verification methods aim to provide certified guarantees that no adversarial examples exist within a specified region of the input space. While these approaches offer stronger assurances, they currently scale only to restricted network architectures or low-dimensional inputs [Wur24]. As a result, heuristic verification remains indispensable in practice, but its limitations must be clearly acknowledged and mitigated through careful experimental design and threat-model-consistent evaluation [CW17].

3.1 Evaluating Adversarial Robustness

Evaluating adversarial robustness is fundamentally challenging because robustness is defined with respect to a specific threat model and a class of admissible perturbations. A naive evaluation strategy applies a small set of standard attacks and concludes that a model is robust if these attacks fail. As discussed in the previous section, this approach is unreliable and can lead to overly optimistic robustness claims [Cha+21; Bas+17].

One major problem of naive evaluation is the use of attacks that are too weak or insufficiently adapted to the defense. Single-step or heuristic gradient-based attacks may fail even when adversarial examples exist, especially in the presence of gradient masking or numerical instabilities [CW17]. In such cases, the failure of an attack reflects a limitation of the evaluation method rather than genuine robustness of the model.

Carlini and Wagner [CW17] emphasize that robustness evaluation must be treated as an adversarial process itself [CW17]. When a defense modifies the optimization landscape, the attacker's objective function, optimization strategy, or internal representation being targeted must be adapted accordingly. In particular, attacks should operate on logits rather than post-softmax probabilities, avoid assumptions about gradient smoothness, and explicitly optimize perturbation size under the chosen L_p constraint [CW17].

Another key recommendation is to evaluate robustness under multiple distance metrics. Since different L_p norms capture different types of perturbations, a model may appear robust under one norm while being highly vulnerable under another [Cha+21]. Consequently, restricting evaluation to a single metric can give a misleading picture of overall robustness.

Finally, robust evaluation requires transparent reporting of failure modes. This includes not only attack success rates but also achieved perturbation magnitudes, confidence levels of misclassification, and cases where optimization fails to converge [CW17]. Reporting such details helps distinguish genuine robustness from artifacts caused by obfuscated gradients or numerical issues. Reliable adversarial robustness evaluation requires strong, adaptive, and optimization-based attacks, consideration of multiple threat models and distance metrics, and careful reporting of experimental outcomes. Without these precautions, heuristic evaluation risks overstating the security of neural network models [CW17; Cha+21].

4 Core Paper Analysis

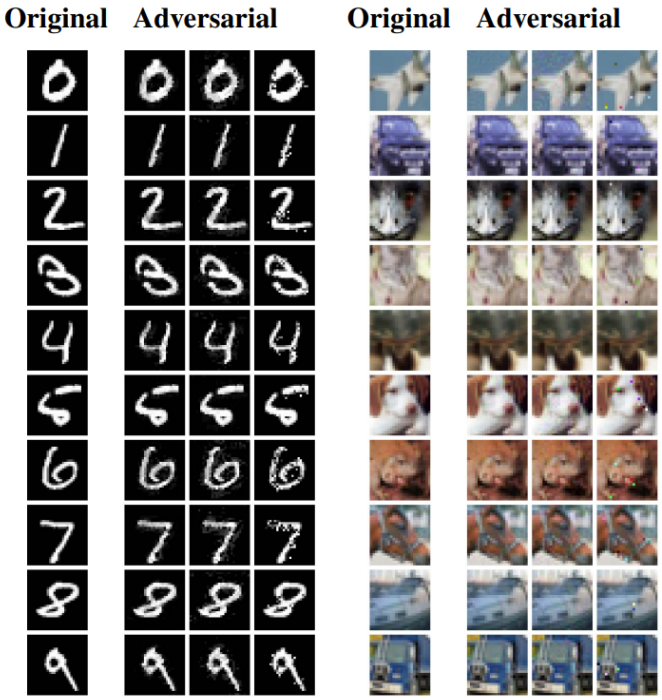


Figure 2: Reprinted from N. Carlini and D. Wagner, Toward Evaluating the Robustness of Neural Networks, 2017 [CW17]. Examples adversarial Attacks on MNIST(left) and CIFAR-10(right), where the perturbations are so small that changes are barely visible to the human eye. The three adversial examples were created from left to right using L_2, L_∞ and L_0 algorithms.

4.1 Problem Setting and Evaluation Benchmarks

In Carlini and Wagner [CW17], three new attack algorithms demonstrate that defensive distillation fails to provide genuine robustness against adaptive white-box attacks. The authors evaluate these attacks on standard vision benchmarks: MNIST, CIFAR-10, and ImageNet. MNIST consists of 28×28 grayscale handwritten digits and serves as a simple testbed with well-understood decision boundaries (see Fig. 3). CIFAR-10 contains 32×32 color images from ten object classes, posing greater challenges due to higher variability and dimensionality (see Fig. 3). ImageNet finally comprises millions of high-resolution color images across 1000 classes, representing real-world complexity that current defenses struggle to handle. ImageNet and ImageNet-like models further scale

this setting to high-resolution images and a thousand or more object categories, which reflects the complexity of modern deep learning applications in computer vision. Carlini and Wagner [CW17] study the robustness of image classification models against targeted, white-box adversarial attacks under different L_p-norms. The primary task is to find, for a given input x and a target class t, the smallest possible perturbation δ such that the perturbed input $x' = x + \delta$ is classified as t, where the perturbation δ remains within the chosen distance metric, i.e., the L_p-norm.

To make their results comparable to prior work and to cover different levels of complexity, the authors evaluate their attacks on the 3 vision benchmarks. Using this spectrum of datasets allows the authors to demonstrate that their attacks are not limited to toy problems, but scale from small to large models and from low- to high-dimensional input spaces.

The choice of benchmarks is closely tied to the threat models discussed in the background section. On MNIST and CIFAR-10, robustness is typically measured with respect to L_∞ or L_2 -bounded perturbations, which model small but widespread pixel-wise changes and global noise. On ImageNet-scale models, the same norms are used, but the much higher dimensionality makes it particularly challenging to ensure that an attack truly finds near-minimal perturbations rather than getting stuck in local optima. By successfully attacking defensively distilled networks across MNIST, CIFAR-10, and ImageNet with three new algorithms, Carlini and Wagner [CW17] demonstrate that distillation provides only apparent robustness—primarily through gradient masking that misleads weaker attacks like FGSM and JSMA, rather than eliminating adversarial examples. This underscores the need for scalable, adaptive attacks that provide reliable upper bounds on model robustness, distinguishing genuine defenses from security through obscurity. In contrast, the proposed optimization-based attacks are explicitly adapted to the distorted loss landscape and remain highly effective even on defensively distilled models, achieving near-100% targeted success rates under L_0, L_2 and L_∞ constraints. This contrast shows that defensive distillation mainly induces gradient masking and that robustness claims based solely on the failure of weak or non-adaptive attacks are unreliable [CW17; Car22].

4.2 Attacks under Different Norms

Attacker Goals

Adversarial threats are defined with respect to the objective that must be protected. For learning-based classifiers, the primary security objective is typically the *integrity* of the prediction: the system should output the correct label for legitimate inputs at inference time. However, depending on the application, adversaries may also target *availability* (e.g., causing systematic failures or denial of service) or *confidentiality* (e.g., extracting sensitive information about training data or model parameters) [Cha+21]. In the setting of adversarial examples studied in this report, we focus on *evasion attacks* against integrity at inference time, where the attacker supplies a modified input x' that induces an incorrect prediction.

Concretely, an attacker's goal in evasion can take different forms [SPV19]: (i) *confidence reduction*, where the attacker aims to reduce the classifier's confidence in the correct class and thereby induce ambiguity; (ii) *untargeted (random) misclassification*, where any label different from the true class is acceptable; and (iii) *targeted misclassification*, where the attacker enforces a specific target class t. In adversarial example research, the most common benchmark is targeted (or at least explicitly specified) misclassification, because it requires more control over the decision boundary and yields a stricter robustness stress test.

The L_2 Attack

The L_2 version of the Carlini and Wagner [CW17] attack formulates adversarial example generation as the following optimization problem:

$$\min_w \left\| \frac{1}{2}(\tanh(w)+1) - x \right\|_2^2 + c \cdot f\left(\frac{1}{2}(\tanh(w)+1)\right),$$

where

$$f(x') = \max\Big(\max\{Z(x')_t : i \neq t\} - Z(x')_t,\ -\kappa\Big).$$

Here, x denotes the original input, $x' = \frac{1}{2}(\tanh(w) + 1)$ is the adversarial example obtained by optimizing over the unconstrained variable w, and $Z(x')$ are the logits of the network. The $\tanh$ reparameterization ensures that x' stays within the valid input range $[0, 1]^n$. The parameter $c > 0$ trades off between minimizing the L_2 distortion and enforcing misclassifications, while the confidence parameter $\kappa \geq 0$ controls how strongly the target logit $Z(x')_t$ must dominate all other logits. Increasing κ forces the optimizer to find adversarial examples that are classified as the target class t with higher confidence, typically at the cost of larger perturbations. In practice, c is chosen via a binary search and the objective is minimized using gradient-based optimization (e.g., Adam) [CW17].

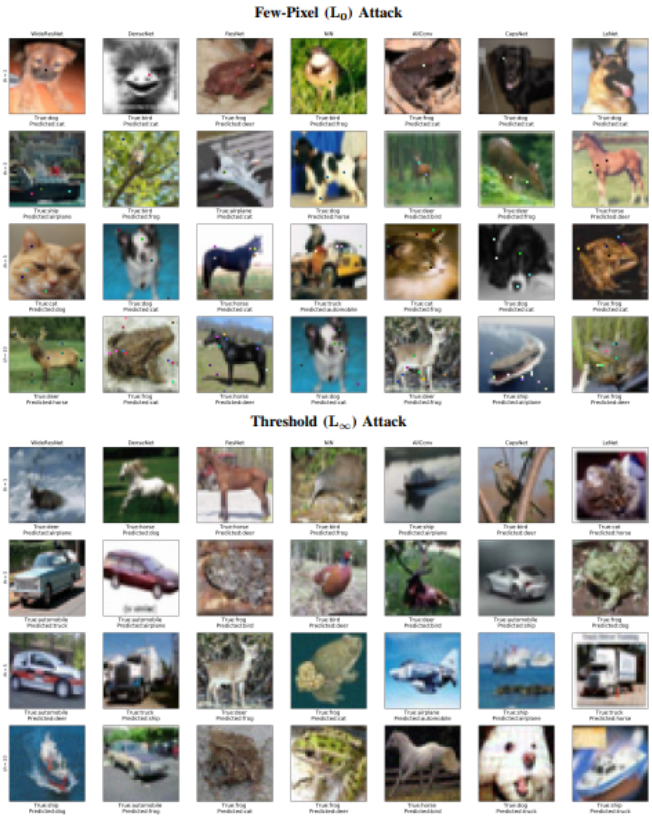


Figure 3: Reprinted from N. Carlini and D. Wagner, Toward Evaluating the Robustness of Neural Networks, 2017 [CW17]. Adversarial samples found with Few-pixel (L_0) black-box attack and threshold (L_1) black-box attack.

The L_0 Attack

As mentioned in the background, the L_0 "norm" is not differentiable and therefore not directly amenable to gradient-based optimization [CW17]. For this reason, Carlini and Wagner [CW17] propose an iterative procedure that repeatedly solves an L_2 attack and gradually reduces the set of pixels that may be modified. The goal of this procedure is to find an adversarial example that changes as few pixels as possible, i.e., to approximate a minimally sparse L_0 perturbation. In each iteration, an L_2 attack is run under the current set of allowed pixels, and the pixels that contribute least to the classifier's output are identified and removed from this set. This importance is estimated by measuring how the loss or the logits change when individual pixels are fixed or masked, so that pixels with minimal influence are discarded first. The process is repeated until the L_2 attack can no longer find a successful adversarial example, so that the remaining pixels constitute a minimal subset whose manipulation is necessary to change the prediction. The trade-off parameter c

is initialized with a small value and then increased, typically by doubling, until either a successful adversarial example is found or a predefined upper threshold is reached. This L_0 attack is particularly suited to threat models in which the adversary can only modify a small number of input dimensions, for example when only a few pixels or features can realistically be changed [CW17]. However, the iterative elimination procedure is computationally much more expensive than a single L_2 attack, since many L_2 optimization problems have to be solved in sequence. In practice, the resulting adversarial examples tend to be very sparse but often visually more localized or noticeable than the more diffuse perturbations produced by the L_2 attack [CW17].

The L_∞ Attack

The L_∞ variant of the Carlini and Wagner [CW17] attack aims to find an adversarial example whose maximum per-pixel change is as small as possible, while still enforcing targeted misclassification. Instead of directly constraining $\|\delta\|_\infty$, Carlini and Wagner [CW17] introduce an auxiliary threshold parameter τ and consider the following objective:

$$\min \; c \cdot f(x+\delta) + \sum_i \left[(|\delta_i| - \tau)^+\right].$$

where $(z)^+ = \max(z, 0)$ denotes the positive part, and $f(\cdot)$ is the same attack loss used in the L_2 formulation. The penalty term $\sum_i (|\delta_i| - \tau)^+$ encourages all pixel-wise perturbations δ_i to stay within the interval $[-\tau, \tau]$; only deviations larger than τ are penalized. The algorithm proceeds by fixing a value of τ and minimizing this objective using gradient-based optimization, typically with the same solver as in the L_2 attack. If a successful adversarial example is found such that $|\delta_i| \leq \tau$ for all i, the threshold τ is reduced (e.g., multiplied by a factor of 0.9) and the optimization is repeated. Otherwise, if no adversarial example can be obtained for the current τ, the search is terminated and the last successful solution is returned. The trade-off parameter c is chosen as in the L_2 attack. This iterative tightening of τ effectively searches for the smallest L_∞ radius within which a targeted adversarial example exists, making the attack conceptually similar to a bisection on the robustness radius. In practice, the L_∞ attack has comparable runtime to the L_2 version, since each step still relies on standard gradient-based optimization, but it aligns directly with the widely used L_∞ threat model in adversarial training and robustness benchmarks [CW17].

4.3 Attack Algorithms

While the previous section introduced adversarial attacks from the perspective of different L_p threat models, these formulations do not by themselves specify how adversarial examples are computed in practice. Concrete attack algorithms provide algorithmic instantiations of these threat models by defining specific optimization strategies, approximations, or heuristics for constructing adversarial perturbations under a given norm constraint.

Early adversarial attacks established the foundational paradigms for generating misclassifying perturbations but were limited by heuristic approximations, specific norm assumptions, and poor scalability to defended models. This subsection reviews key historical algorithms: L-BFGS, FGSM, and JSMA, categorized by their primary L_p -norm threat models, highlighting how they paved the way for the more systematic, optimization-based framework of Carlini and Wagner [CW17]. While L-BFGS optimized minimal L_2 distortion via constrained solvers, FGSM enabled fast L_∞-bounded single-step attacks, and JSMA targeted sparse L_0 changes through saliency analysis, these methods often failed against gradient masking or distillation due to reliance on linear approximations or repeated Jacobians. In contrast, Carlini and Wagner [CW17] generalize across norms with explicit minimization of perturbation size under a unified objective, providing stronger adaptive attacks for reliable robustness evaluation.

L-BFGS Attack

The first systematic construction of adversarial examples was introduced by Szegedy et al. [Sze+13], who formulated the attack as a constrained optimization problem solved with the L-BFGS algorithm. Given an input x with the true label y, the goal is to find a small perturbation δ such that

$x + \delta$ is misclassified L_2 distortion $\|\delta\|_2$ as small as possible Carlini and Wagner [CW17]. This formulation provided an early proof-of-concept that adversarial examples are not isolated artifacts but can be generated reliably by numerical optimization Carlini and Wagner [CW17]. However, the method requires solving a separate, relatively costly optimization problem for each input, which makes it difficult to scale to modern deep architectures and large datasets. Consequently, the L-BFGS attack is now mainly of historical importance: it established the optimization-based view on adversarial example generation and paved the way for more efficient gradient-based attacks that are better suited for large-scale robustness evaluation Carlini and Wagner [CW17].

Fast Gradient Sign Method (FGSM)
The Fast Gradient Sign Method, proposed by Goodfellow, Shlens, and Szegedy [GSS15], is a simple and efficient one-step gradient-based attack defined under an L_∞ threat model. FGSM perturbs the input in the direction of the sign of the gradient of the loss with respect to the input:

$$x' = x + \varepsilon \cdot \text{sign}(\nabla_x \mathcal{L}(x, y)).$$

Here, ε controls the maximum per-coordinate perturbation. FGSM is computationally cheap and therefore widely used, but its reliance on a single linear approximation makes it relatively weak and susceptible to gradient masking and defensive techniques [CW17].

Jacobian-based Saliency Map Attack (JSMA)
The Jacobian-based Saliency Map Attack, introduced by Papernot et al. [Pap+16], is a targeted attack that aims to modify as few input features as possible and is therefore commonly associated with an L_0 threat model. JSMA computes the Jacobian of the network outputs with respect to the input features and constructs a saliency map that identifies which features most strongly increase the target class score while simultaneously decreasing the scores of competing classes. The attack then iteratively modifies the most salient features (often in pairs) until the target classification is achieved. While JSMA can produce very sparse adversarial examples, it is computationally expensive due to the repeated Jacobian and saliency-map computations and scales poorly to high-dimensional inputs.

Overall, while these attack algorithms significantly advanced the understanding of adversarial vulnerability, their heuristic nature and limitations under defended models motivated the optimization-based approach adopted by Carlini and Wagner [CW17], which explicitly minimizes perturbation size and remains effective against gradient obfuscation [CW17]. In particular, Carlini and Wagner [CW17] reinterpret adversarial example generation as a flexible optimization framework that can be instantiated under different L_p threat models and tuned to reliably attack even gradient-masked or defensively distilled networks, thereby providing a stronger and more systematic basis for robustness evaluation than earlier, single-step or Jacobian-based methods [CW17].

4.4 Attack Formulation as Optimization Problem

A central contribution of Carlini and Wagner [CW17] is a unified optimization-based framework for generating adversarial examples. Instead of relying on heuristic gradient updates, the authors explicitly formulate adversarial example generation as the minimization of a well-defined objective function under validity constraints on the input. This general formulation forms the basis for all three attacks proposed in the paper and is later instantiated for different L_p norms.

Objective Function
Let $x \in [0, 1]^n$ be a correctly classified input with label $y = C(x)$, and let $t \neq y$ be a target class. The goal of a targeted attack is to find an adversarial example x' that is classified as t while remaining close to x. Carlini and Wagner [CW17] encode this goal by minimizing a composite loss that balances perturbation size and attack success [CW17].

To avoid numerical issues associated with softmax probabilities, the attack operates directly on the logits $Z(x')$ of the network. The misclassification requirement is captured by a differentiable

loss function $f(x')$ such that $f(x') \leq 0$ implies successful classification as the target class. A standard choice used throughout the paper is

$$f(x') = \max\Big(\max_{i \neq t} Z(x')_i - Z(x')_t,\ -\kappa\Big),$$

where $\kappa \geq 0$ is a confidence parameter. Larger values of κ enforce that the logit of the target class exceeds all competing logits by a larger margin, producing higher-confidence adversarial examples at the cost of increased distortion [CW17].

The overall optimization problem takes the general form

$$\min_{x'}\ \|x' - x\|_p + c \cdot f(x'),$$

where $\|\cdot\|_p$ denotes the chosen distance metric and $c > 0$ is a trade-off parameter that controls the balance between minimizing perturbation size and enforcing misclassification. In practice, c is selected via a binary search to obtain adversarial examples with minimal distortion that still satisfy the attack constraint [CW17].

Box Constraints

In image classification tasks, inputs are typically constrained to a bounded domain such as $[0, 1]^n$. Directly optimizing over x' with such box constraints is inconvenient for gradient-based methods, as it requires repeated projection or clipping steps that can lead to unstable gradients.

To address this issue, Carlini and Wagner [CW17] introduce a reparameterization that transforms the constrained optimization problem into an unconstrained one. Specifically, the adversarial example is expressed as

$$x' = \frac{1}{2}(\tanh(w) + 1),$$

where $w \in \mathbb{R}^n$ is an unconstrained optimization variable. This transformation guarantees that x' always lies in the valid input range $[0, 1]^n$ and removes the need for explicit projection operations during optimization [CW17].

As a result, the full objective becomes smooth and unconstrained in w, allowing the use of standard gradient-based optimizers such as Adam. This reparameterization is crucial for the stability and effectiveness of the attack and plays an important role in bypassing defenses that rely on gradient obfuscation or numerical instabilities [CW17].

4.5 Evaluation Results and Implications

Carlini and Wagner [CW17] revisit defensive distillation to test whether it yields genuine robustness or merely obscures standard attack methods. Their experiments on MNIST and CIFAR-10 show that distilled models remain vulnerable to targeted, white-box adversarial examples under multiple L_p norms. While earlier gradient-based attacks often fail against distilled models, the proposed optimization-based attacks reliably find adversarial examples with small distortion under L_0, L_2, and L_∞ threat models [CW17]. This demonstrates that adversarial examples are not removed by distillation; instead, the defense primarily changes the loss landscape in a way that makes certain attacks ineffective.

A central implication is that robustness evaluations based only on a limited set of attacks can be misleading. Defensive distillation is a paradigmatic case where a defense appears successful under non-adaptive, heuristic attacks, yet fails under stronger adaptive optimization procedures [CW17]. In modern terminology, this behavior is consistent with *gradient masking*: gradients become numerically unhelpful for some attack formulations, which creates a false impression of robustness even though adversarial examples still exist [CW17]. More generally, surveys of adversarial machine learning emphasize that many defenses exhibit this pattern: they provide apparent robustness against specific attacks but do not generalize to stronger or adapted adversaries [Cha+21].

Another important lesson concerns the role of the chosen distance metric. Carlini and Wagner [CW17] explicitly study multiple norms and show that robustness conclusions depend on whether perturbations are constrained by sparsity (L_0), overall magnitude (L_2), or per-coordinate

bounds (L_∞). Because these norms correspond to different practical attack capabilities, evaluation restricted to a single norm can overstate security and miss relevant vulnerabilities [CW17; KV20]. Consequently, the threat model must be made explicit, and evaluation should ideally cover multiple norms to obtain a more complete view of model vulnerability [Cha+21].

Carlini and Wagner [CW17] clarify a methodological point: attack-based evaluation typically provides an *upper bound* on robustness. A successful attack certifies non-robustness beyond the found distortion, but failure of an attack is not evidence of robustness unless stronger guarantees are available. This aligns with the broader distinction between heuristic robustness evaluation and certified verification approaches, where only the latter can provide lower bounds (guarantees of robustness within a region), though often at limited scale [Bas+17; Wur24]. Taken together, the results of Carlini and Wagner [CW17] imply that defensive distillation does not substantially increase true adversarial robustness and that reliable robustness claims require adaptive, strong attacks and transparent, threat-model-consistent evaluation across relevant perturbation models [CW17; Cha+21]. These conclusions are drawn under white-box, digital L_p threat models; robustness against other, more realistic or physical attacks is not evaluated in Carlini and Wagner [CW17].

5 Related Work

This section positions our work within the broader literature by contrasting empirical attack- and defense-based evaluations with recent approaches to formal robustness verification. Rather than exhaustively cataloguing individual methods, we focus on the conceptual shift from heuristic security assessments toward principled, verifiable robustness guarantees. This perspective motivates the transition from survey-based analyses to formal verification methods discussed below.

Early analyses of adversarial robustness revealed a fundamental mismatch between the perceived and the actual security of neural networks. Among the most influential contributions, Carlini and Wagner [CW17] systematically exposed the weakness of contemporaneous defenses and demonstrated that even small, carefully optimized perturbations could reliably fool state-of-the-art models. This realization marked a turning point, shifting the community's attention toward developing rigorous evaluation methods and principled robustness assessments. Adversarial attacks and robust defense strategies have become a central focus of research in the field of deep learning. Numerous review papers structure the diversity of existing attack and defense mechanisms, while newer approaches increasingly strive for formal robustness guarantees. In the following, survey papers on adversarial attacks and defenses are first considered before formal methods for robustness verification are introduced. This allows us to understand the transition from purely empirical security considerations to formal verification.

Previous work first provides a broad overview of adversarial attacks and defense mechanisms in deep learning systems. *A Survey on Adversarial Attacks and Defenses* structures the research landscape by classifying attacks according to access type (white-/black-box), objective (evasion vs. poisoning), attack phase (training vs. inference), and perturbation model (e.g., L_p-bounded vs. semantic transformations), and systematically compares defense approaches such as adversarial training, input preprocessing, detection, and model-based methods. The survey emphasizes recurring pitfalls in evaluation, including gradient masking, obfuscated gradients, and overfitting to a single attack, and argues for standardized benchmarks, adaptive threat models, and stronger evaluation protocols that go beyond testing against a fixed, non-adaptive attacker [Cha+21]. In particular, it highlights that many defenses that initially report strong robustness on MNIST or CIFAR-10 lose most of their purported protection once the attacker is allowed to adapt to the defense and stronger optimization-based or multi-step attacks are applied [Cha+21].

In addition, *Adversarial Attacks and Defenses in Deep Learning: From a Perspective of Cybersecurity* explicitly examines the same issue from a security-oriented perspective. The focus here is on concrete real-world threat scenarios, such as attacks on payment systems, biometric authentication, autonomous vehicles, or industrial control systems, and how adversarial examples can be weaponized in these domains [Zho+22]. Beyond cataloguing known defense mechanisms, the survey discusses practical aspects such as runtime overhead, deployment complexity, compatibility with legacy systems, and the trade-off between robustness and clean accuracy, and stresses that many academic defenses are difficult to integrate into production-grade cybersecurity environments [Zho+22]. Both surveys make it clear that classic defense strategies such as simple input

denoising, gradient masking, or non-adaptive adversarial training are often insufficient to ensure long-term resilience against adaptive attacks in security-critical applications, and they call for defense designs that are explicitly grounded in well-defined threat models, accompanied by rigorous, reproducible evaluation standards and, where possible, certified robustness guarantees [Zho+22; Cha+21].

While surveys primarily classify empirical defense mechanisms and attack techniques, more recent work increasingly addresses the question of how robustness can be formally verified, i.e., with mathematical guarantees. This is where the paper Robustness Verification in Neural Networks comes in, shifting the focus from the construction of ever-new defenses to the systematic verification of the robustness of existing models.

Robustness verification in neural networks investigates methods that provide formal guarantees for the robustness of neural networks against certain classes of adversarial perturbations [Wur24]. Formal methods such as solver-based approaches (e.g., SAT/SMT) as well as linearizations and relaxations are used to verify that a model does not exhibit misclassifications in a given input range [Wur24]. Compared to purely empirical defenses, this approach enables the generation of proofs or counterexamples, thereby explicitly addressing the gap identified in the surveys regarding the lack of reliable robustness guarantees [Wur24]. At the same time, it becomes clear that scalability to large, realistic networks remains a challenge, making heuristic and approximate verification methods necessary [Wur24]. Wurm [Wur24] frames these questions in a unified complexity-theoretic setting and studies several decision problems, such as whether all inputs in a given polyhedral region are mapped to safe outputs or whether two networks compute the same function [Wur24]. For piecewise linear activation functions and sum- or max-based distance metrics, many of these robustness and equivalence problems turn out to be decidable in polynomial time or at most NP, which shows that they are in principle tractable in a semi-linear setting [Wur24]. At the same time, the paper highlights that exact verification quickly becomes computationally expensive once network size and input dimension grow, which explains why practical tools often rely on mixed-integer formulations or convex relaxations that trade completeness for efficiency [Wur24]. This theoretical perspective complements existing empirical robustness evaluations by clarifying which guarantees are algorithmically feasible and where fundamental complexity barriers arise [Wur24]. Consequently, robustness verification is positioned as a bridge between heuristic attack-based evaluation and fully certified safety guarantees for neural network models in safety-critical applications [Wur24].

In summary, the studies reviewed show that the sheer diversity of adversarial attacks and defenses is not sufficient to ensure robustness in security-critical applications [Wur24; Zho+22; Cha+21]. Surveys identify open problems such as the lack of standard benchmarks and the circumvent ability of many defenses, while formal verification approaches point to initial ways of mathematically substantiating robustness guarantees, but still suffer from scalability and model assumptions.

6 Discussion and Conclusion

The central insight is that robustness claims are only meaningful relative to a clearly specified threat model and a sufficiently strong evaluation procedure. In particular, Carlini and Wagner [CW17] demonstrate that defensive distillation does not yield genuine robustness: the defense mainly changes the optimization landscape such that standard gradient-based attacks become unreliable, while adaptive optimization-based attacks still succeed under L_0, L_2, and L_∞ constraints. This establishes gradient masking as a key pitfall in robustness evaluation and shows that the failure of a specific attack is not evidence of robustness. However, despite its title, the work does not propose a unified metric or framework for robustness, but rather a set of strong optimization-based attacks that provide empirical upper bounds under L_p -bounded threat models Carlini et al. [Car+19]. This focus on norm-constrained, pixel-level perturbations leaves other practically relevant threat models, such as semantic or physical attacks, largely unaddressed, limiting the scope of the robustness claims Carlini et al. [Car+19] and Carlini [Car22].

A second implication concerns the relationship between attacks and defenses. Many early defenses were evaluated against a narrow class of attacks and appeared effective largely because they

exploited weaknesses of these attacks. In contrast, the C&W framework treats adversarial example generation as an optimization problem and thereby produces strong, threat-model-consistent attacks that serve as practical upper bounds on robustness. From a broader perspective, surveys confirm that a large fraction of defenses exhibit limited generalization and can often be circumvented once the attacker adapts to the defense mechanism [Cha+21; Zho+22]. Consequently, robust machine learning requires not only proposing defenses but also establishing evaluation protocols that are resistant to such adaptive adversaries. Threat models play a decisive role in how robustness results should be interpreted. Throughout this work, we primarily considered the white-box setting, where the adversary has full knowledge of the model and its parameters, and where optimization-based attacks represent a strong and realistic worst-case evaluation tool [CW17]. However, many practical attack scenarios are black-box, where adversaries rely on transferability or query access. Prior work has shown that adversarial examples can transfer across models, enabling black-box attacks even without gradient access [Sze+13; GSS15]. In addition, physical-world attacks introduce further constraints such as viewpoint changes, illumination shifts, and sensor noise, which are not fully captured by standard L_p perturbation models in pixel space [Cha+21]. Therefore, robustness conclusions derived under purely white-box, digital L_p threat models do not automatically generalize to all deployment settings, and selecting an appropriate perturbation model remains an application-dependent design choice [KV20]. In retrospect, even the authors said that the paper should be read primarily as a step towards stronger attack-based evaluation, not as a comprehensive recipe for robustness, and later work explicitly broadens the discussion to more diverse threat models and methodological best practices Carlini et al. [Car+19] and Carlini [Car22]. Several open questions remain. First, there is still no consensus on defenses that reliably generalize across attack families and threat models. Existing defenses often trade robustness for clean accuracy, suggesting an inherent tension between standard generalization performance and worst-case robustness [SHS19].

Second, there is increasing interest in moving beyond purely heuristic evaluation toward certified guarantees. Formal verification and certification methods can provide lower bounds on robustness, but scalability remains a key challenge for modern, large-scale architectures [Bas+17; Wur24].

Third, theoretical work suggests that adversarial vulnerability is closely related to high input dimensionality and local linear behavior, indicating that robustness may not be eliminated by simple fixes but may require principled changes in model design, training objectives, or data representations [Sim+19; GSS15].

In conclusion, Carlini and Wagner [CW17] provide a key methodological lesson for robust machine learning: robustness is not established by defenses alone, but by rigorous, adaptive, and threat-model-consistent evaluation. Their work remains influential because it exposes how easily robustness can be overestimated through heuristic verification and because it motivates both stronger evaluation protocols and the pursuit of defenses and verification methods that provide robustness beyond specific attack heuristics [CW17; Cha+21].

Statement of Contributions

Contributions by Clara Rittmüller:

- Introduction
- Background
- Problem Setting and Evaluation Benchmarks
- Attacks under Different Norms
- Related Work

Contributions by Arif Onur Alp:

- Heuristic Verification
- Attacks under Different Norms
- Attack Algorithms
- Attack Formulation as Optimization Problem
- Discussion and Conclusion

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